

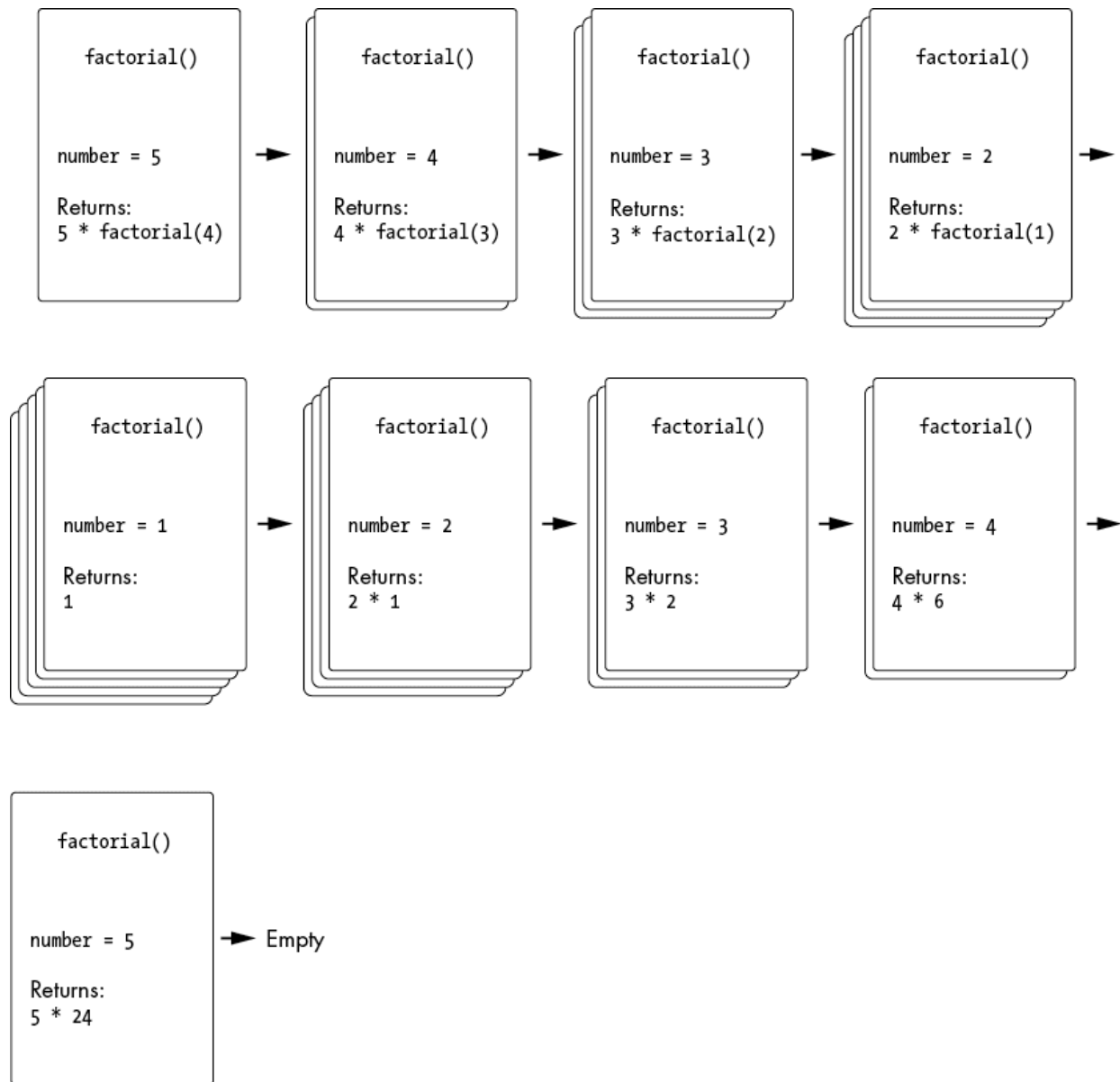


Figure 2-1: The state of the call stack as the recursive calls to `factorial()` are called and then return

The iterative factorial algorithm, on the other hand, will complete the calculation quickly and efficiently. The stack overflow can be avoided using a technique available in some programming languages called *tail call optimization*. Chapter 8 covers this topic. However, this technique further complicates the implementation of the recursive function. For calculating factorials, the iterative approach is the simplest and most direct.